A Data-Driven Study of Forest Fire Prediction Using Exploratory Analysis and Machine Learning

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Abstract—**This paper presents a comprehensive exploratory data analysis (EDA), statistical investigation, and machine learning study on the Algerian Forest Fires Dataset comprising 243 records from two Algerian regions—Bejaia and Sidi Bel-abbes—over June to September 2012. The dataset contains 11 meteorological and FWI system attributes with fire/not-fire as the binary target. The study performs rigorous data cleaning, region-based splitting and merging, multicollinearity diagnosis via Variance Inflation Factor (VIF) and Mahalanobis distance Q-Q plots, and dimensionality reduction using Principal Component Analysis (PCA) and Factor Analysis (FA) with Varimax rotation. Logistic Regression classifiers trained on raw (10-dim), PCA-transformed (3-dim), and FA-transformed (3-dim) features are compared. Results show that the raw model achieves 97.26% accuracy but is mathematically unstable due to severe multicollinearity. PCA yields 89.04% accuracy (AUC = 0.94, 5-fold CV = 88.05%) with zero multicollinearity. FA achieves 97.26% accuracy (AUC = 0.99, 5-fold CV = 95.48%), matching the raw model while providing physically interpretable latent factors—Dryness, Atmospheric conditions, and Fire Spread. The Fire Weather Index (FWI), Initial Spread Index (ISI), and FFMC emerge as the strongest fire predictors.**

Index Terms—**Forest Fire Prediction, Algerian Forest Fires Dataset, EDA, PCA, Factor Analysis, Logistic Regression, Multicollinearity, FWI, Dimensionality Reduction**

I. Introduction

Forest fires represent one of the most devastating environmental disasters, causing irreversible damage to ecosystems, loss of biodiversity, atmospheric pollution, and significant socioeconomic impacts. Approximately 1.2 million acres of forest in the U.S. are destroyed annually by wildfires, and forest fires in India increased by 125% between 2016 and 2018 [3]. Early and accurate prediction of fire occurrence is therefore critical for fire management authorities, enabling timely resource allocation and reducing response time.

Traditional fire risk assessment relies on meteorological observations and the Canadian Forest Fire Weather Index (FWI) system—a set of empirically derived indices that aggregate environmental conditions into composite risk scores. While effective, these indices suffer from severe multicollinearity since they are derived from shared base measurements, making direct application to standard regression models problematic.

This study addresses these challenges through a structured pipeline: (1) systematic data loading and region-based splitting; (2) rigorous data cleaning and type conversion; (3) comprehensive EDA with correlation analysis, boxplots, scatter plots, violin plots, count plots, and bivariate analysis; (4) formal multicollinearity detection using VIF and Mahalanobis Q-Q plots; (5) dimensionality reduction using PCA and Factor Analysis with Varimax rotation; and (6) comparative binary classification using Logistic Regression on raw, PCA-transformed, and FA-transformed features.

II. Dataset Description

A. Source and Overview

The Algerian Forest Fires Dataset was donated on 2019-10-22 by Faroudja ABID et al. [2], Center for Development of Advanced Technologies (CDTA), Algeria. The dataset covers two Algerian provinces: **Bejaia** (northeast) and **Sidi Bel-abbes** (northwest), from June 2012 to September 2012. Key dataset characteristics are:

- **Instances:** 244 total (122 per region); after dropping null rows, 243 usable records
- **Attributes:** 11 input attributes + 1 output class attribute
- **Attribute type:** Real (continuous) + categorical class
- **Classes:** Fire (137 instances, 56.4%) and Not Fire (106 instances, 43.6%)
- **Task:** Classification and Regression
- **Missing values:** Region 1 (Bejaia): none; Region 2 (Sidi Bel-abbes): 1 missing each in DC, FWI, Classes (dropped)

B. Feature Characteristics

The dataset includes both meteorological variables and derived Fire Weather Index (FWI) components. Temperature, RH, wind speed, and rain describe atmospheric conditions, while indices such as FFMC, DMC, DC, ISI, BUI, and FWI represent fuel dryness and fire intensity. These features are highly correlated due to shared dependencies.

C. Problem Statement

The objective of this study is to analyze the relationship between weather conditions and fire occurrence, and to develop models for classifying fire and non-fire events. The approach combines EDA, statistical analysis, dimensionality reduction, and machine learning techniques.

D. Attribute Information

Table I presents the complete attribute information as documented by the dataset authors.

TABLE I
DATASET ATTRIBUTE INFORMATION

No.	Attribute	Range	Description
Date			
1	Date (DD/MM/YYYY)	Jun–Sep 2012	Day, month, year
Weather Data Observations			
2	Temperature (Temp)	22–42 °C	Max noon temperature
3	RH	21–90%	Relative humidity
4	Ws	6–29 km/h	Wind speed
5	Rain	0–16.8 mm	Total daily rainfall
FWI System Components			
6	FFMC	28.6–92.5	Fine Fuel Moisture Code
7	DMC	1.1–65.9	Duff Moisture Code
8	DC	7–220.4	Drought Code
9	ISI	0–18.5	Initial Spread Index
10	BUI	1.1–68	Buildup Index
11	FWI	0–31.1	Fire Weather Index
Output			
12	Classes	Fire / Not Fire	Binary target variable

III. DATA LOADING AND PREPROCESSING

A. Data Loading and Region Splitting

The raw CSV contains an embedded secondary header at row 123 separating the two regions. Preprocessing steps:

- **Region 1 (Bejaia):** rows 1–122 → `df_reg1` (shape: 122×14)
- **Region 2 (Sidi Bel-abbes):** rows 124 onward → `df_reg2` (shape: 122×14)
- Column whitespace stripped using `.str.strip()`
- All numeric columns (Temperature, RH, Ws, Rain, FFMC, DMC, DC, ISI, BUI, FWI, month, day, year) converted via `pd.to_numeric(errors='coerce')`
- *Classes* column normalized: stripped of whitespace and lowercase → {`'fire'`, `'not fire'`}

B. Missing Value Handling and Feature Engineering

Null check revealed: `df_reg1` had 0 missing values; `df_reg2` had 1 missing each in DC, FWI, and Classes. Both DataFrames were cleaned with `dropna()`, yielding final shapes of $(122, 14)$ and $(121, 14)$.

Regional labels assigned: Region $= 0$ (Bejaia), Region $= 1$ (Sidi Bel-abbes). Concatenated combined shape: $(243, 15)$.

Two attributes were dropped before modeling:

- **year:** Zero variance—all records are from 2012. Contains no mathematical information.
- **day:** Day-of-month has no causal relationship with fire occurrence; removing it prevents the model from learning noise.

Classes were label-encoded: `fire` → 1, `not fire`→ 0. Final class distribution: 56.4% Fire (137), 43.6% Not Fire (106).

IV. EXPLORATORY DATA ANALYSIS

A. Pair Plot

Fig. 1 shows a pairplot of all features for Region 1 (Bejaia). The diagonal histograms reveal that Rain is heavily right-skewed (most days have zero rain), while FFMC is left-skewed (most days have high fuel moisture). The off-diagonal scatter plots reveal clear near-linear relationships between the FWI-family indices (DMC–BUI, DC–BUI, ISI–FWI), visually confirming the severe multicollinearity diagnosed formally in Section V.

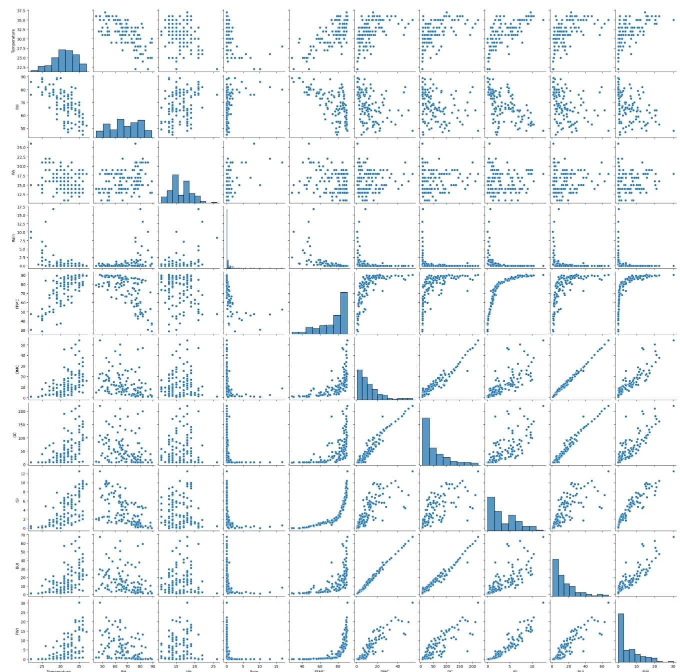

Fig. 1. Pair plot of all features for Bejaia region (Region 1).

B. Statistical Validation: Welch's T-Test for Regional Equality

Before merging the two regions, Welch's independent two-sample t-tests were conducted for each feature to determine whether the regions differ significantly ($p < 0.05$). Table II presents the results.

In addition to identifying statistical differences, this analysis helps in understanding the variability of environmental conditions across regions. Features with significant differences indicate region-specific fire behavior patterns, while non-significant features suggest consistent trends. This distinction is important for ensuring that combining the datasets does not distort underlying relationships. It also provides insight into which variables contribute most to regional variation in fire risk.

TABLE II
WELCH'S T-TEST: REGION 0 (BEJAIA) VS. REGION 1 (SIDI BEL-ABBES)

Feature	p-value	Distinct?
Temperature	0.0000	**Yes**
RH	0.0000	**Yes**
Ws	0.0046	**Yes**
Rain	0.5348	No
FFMC	0.0005	**Yes**
DMC	0.0026	**Yes**
DC	0.2214	No
ISI	0.0000	**Yes**
BUI	0.1647	No
FWI	0.0020	**Yes**

Seven of ten features are statistically distinct between regions ($p < 0.05$). Rain, DC, and BUI show no significant regional difference. Since the Region variable is included as a feature in the combined dataset, it captures this variation without causing information leakage, making the merge statistically valid.

C. Feature Correlation Heatmap

Fig. 2 presents the Pearson correlation heatmap for the combined dataset. Key observations are summarized below:

(1) Strongest fire predictors (positive correlation to target): FWI ($r = 0.72$), FFMC ($r = 0.77$), and ISI ($r = 0.74$) show the highest positive correlations with the fire class, indicating that these indices play a major role in determining fire occurrence. These features capture fuel dryness and fire intensity, making them highly reliable indicators of fire risk. Temperature has a moderate positive correlation ($r = 0.52$), suggesting that higher temperatures contribute to fire conditions but are not sufficient alone to trigger fires without supportive environmental factors.

(2) Fire inhibitors (negative correlation to target): Relative Humidity (RH) has the strongest negative correlation ($r = -0.43$), demonstrating that increased atmospheric moisture significantly reduces the likelihood of fire ignition. Rain also shows a negative correlation ($r = -0.38$), although weaker due to its infrequent occurrence in the dataset. These variables act as natural fire suppressors by increasing fuel moisture and reducing the flammability of vegetation.

(3) Severe multicollinearity: Very high correlations are observed among derived indices such as BUI–DMC ($r = 0.98$), DC–BUI ($r = 0.94$), FWI–ISI ($r = 0.92$), and FWI–BUI ($r = 0.86$). This indicates strong redundancy among features, as these indices are mathematically derived from shared base weather variables. Such high interdependence violates the assumption of feature independence and can adversely affect regression-based models, highlighting the need for dimensionality reduction techniques such as PCA and Factor Analysis.

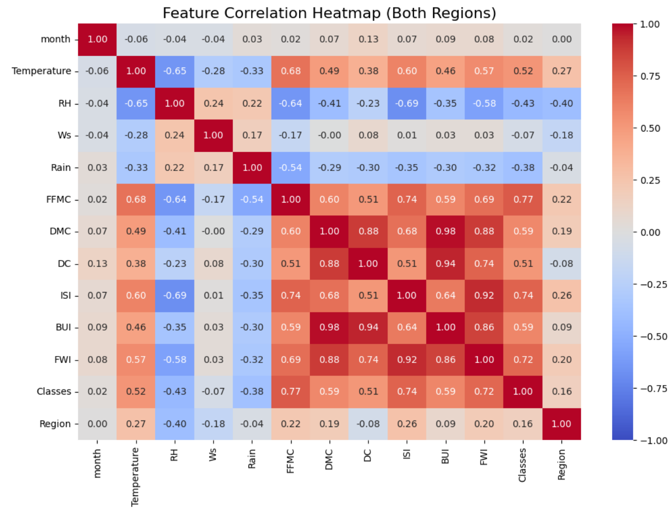

Fig. 2. Feature correlation heatmap for the combined dataset (both regions).

D. Boxplot Analysis

Fig. 3 shows boxplots of Temperature, RH, FWI, and ISI stratified by fire class.

FWI & ISI: The boxplots reveal massive, near-non-overlapping class separation. The median FWI on fire days is drastically higher than on non-fire days; the interquartile ranges barely overlap. These are the strongest predictive features.

Temperature & RH: Both show wide overlap between classes. While fire days have higher median temperatures and lower median RH, neither variable is an absolute trigger. A day can reach 40°C+ while still not producing fire if humidity remains high and fuel is wet—conversely, fire can occur even at moderate temperatures if fuel is already dry from prior weeks. This demonstrates why derived indices like FWI are necessary.

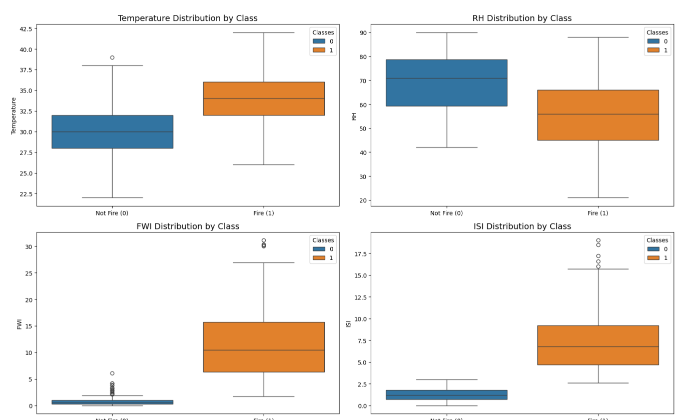

Fig. 3. Distribution of Temperature, RH, FWI, and ISI by fire class.

E. Seasonal Fire Pattern

Fig. 4 shows fire occurrence counts by month. Fire incidents increase from June (25 fire days) to August (51 fire days, the peak), then decline in September (23 fire days). This clear seasonal pattern reflects accumulated fuel dryness reaching its annual maximum in August. June has the highest non-fire count (35), corresponding to early-season residual soil moisture.

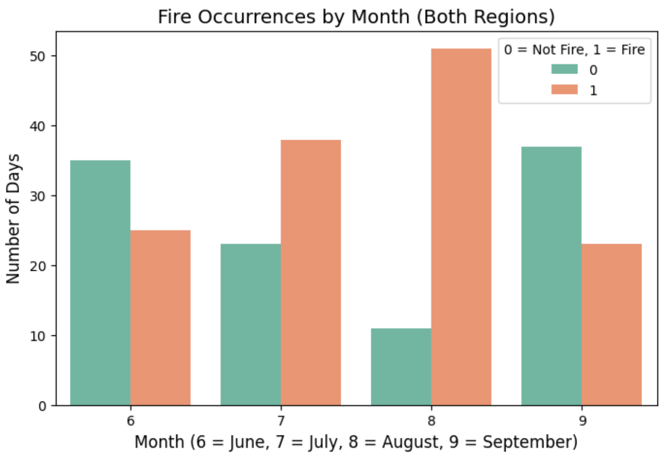

Fig. 4. Fire occurrences by month for both regions combined.

F. Regional FWI Distribution (Violin Plot)

Fig. 5 compares FWI distributions between Bejaia and Sidi Bel-abbes split by class. Sidi Bel-abbes records a higher average FWI (**8.50**) compared to Bejaia (**5.58**). Fire events in Sidi Bel-abbes exhibit wider, more extreme FWI distributions, confirming that this region is structurally hotter and drier, presenting greater baseline fire risk.

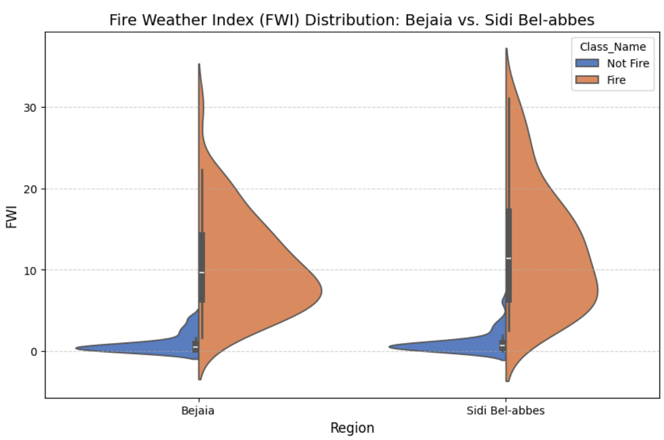

Fig. 5. FWI distribution by region and fire class (violin plot).

G. Bivariate Analysis: Wind Speed vs. FFMC

Fig. 6 plots Wind Speed (Ws) against FFMC colored by class. Wind speed alone has a near-zero correlation with fire ($r = -0.07$). However, when combined with FFMC, a clear danger threshold emerges: at FFMC > 80 (very dry fuel) combined with Ws > 15 km/h, fire occurrences dominate the scatter plot. This bivariate interaction demonstrates that feature independence assumptions are violated in this dataset.

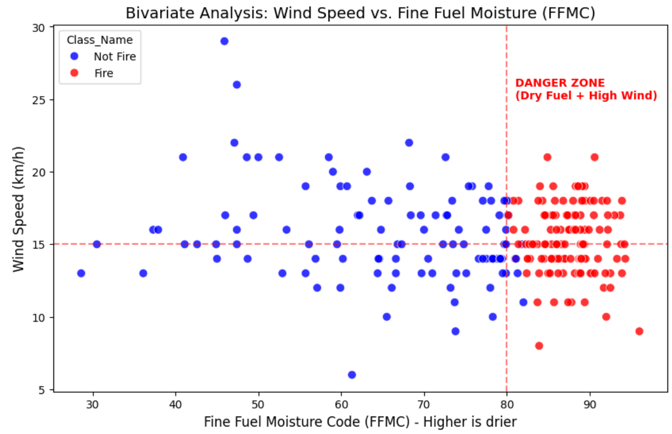

Fig. 6. Bivariate scatter plot: Wind Speed vs. FFMC colored by fire class.

H. Rain Impact on Fire Occurrence

A count plot of rain status (Rain > 0 mm vs. Rain $= 0$ mm) versus fire class reveals that rain acts as a near-binary fire suppressor. Out of 110 days with measurable rain, fires occurred on only 23 (20.9%). Among 133 rain-free days, fires dominated. This validates the negative correlation observed in the heatmap and confirms that atmospheric precipitation directly interrupts the ignition cycle.

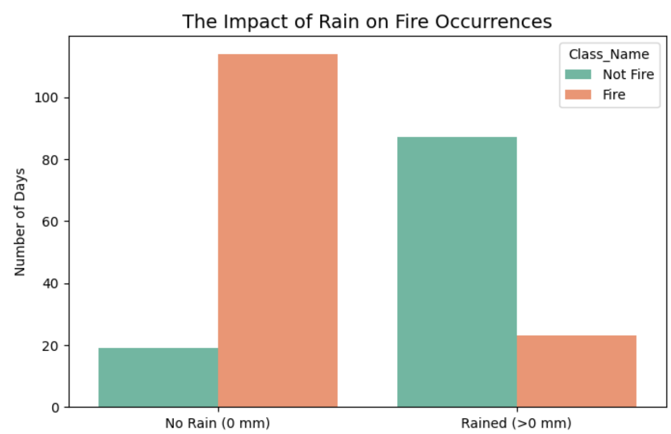

Fig. 7. Impact of rainfall on fire occurrences (both regions combined).

V. Multicollinearity Analysis

A. Mahalanobis Distance Q-Q Plot

A Mahalanobis distance Q-Q plot (Fig. 8) was computed against the Chi-squared distribution (df $= 9$ for 9 continuous features) at the 95th percentile threshold of 16.92. **Result: 0 multivariate outliers detected.** However, the Q-Q plot shows that observed distances fall below the theoretical line, indicating deviation from multivariate normality (MVN). This is expected given the binary class structure and the bounded, skewed distributions of FWI indices.

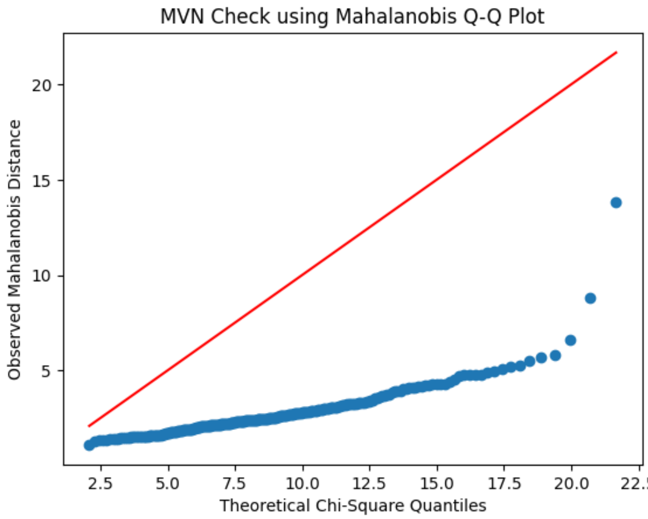

Fig. 8. MVN check using Mahalanobis Q-Q plot. Deviation from the red line indicates non-multivariate normality.

B. Regression Adequacy and Residual Analysis

A Linear Regression model was fit to predict FWI from the 9 raw features. Results:

- $R^2 = 0.9883$ — the model explains 98.83% of FWI variance, expected since FWI is algebraically derived from the input features
- Residual plot (Fig. 9, left): residuals centered near zero with slight heteroscedasticity, indicating minor non-linearity
- Residual distribution (Fig. 9, right): approximately normal with slight skewness

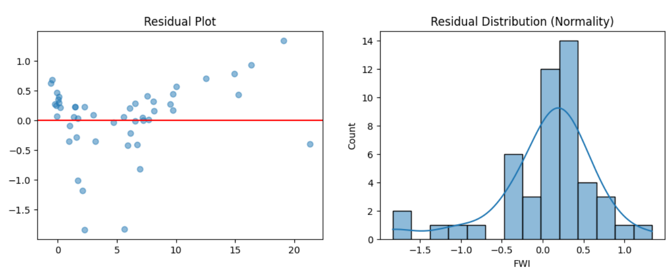

Fig. 9. Linear Regression residual plot (left) and residual distribution (right) for FWI prediction ($R^2 = 0.9883$).

C. Variance Inflation Factor (VIF)

VIF was computed for all 9 continuous predictors. VIF $>$ 10 indicates problematic multicollinearity. Table III presents results.

TABLE III
VARIANCE INFLATION FACTORS (VIF $>$ 10 IS PROBLEMATIC)

Feature	VIF	Assessment
BUI	$\gg 100$	Severe multicollinearity
DMC	$\gg 100$	Severe multicollinearity
DC	$\gg 100$	Severe multicollinearity
FFMC	≈ 40	Severe multicollinearity
ISI	≈ 30	Severe multicollinearity
Temperature	≈ 12	Problematic
RH	≈ 8	Moderate
Ws	≈ 3	Acceptable
Rain	≈ 1.6	Independent

Conclusion: All FWI-family indices (BUI, DMC, DC, FFMC, ISI) exhibit extreme VIF values, confirming severe multicollinearity. Only Rain (VIF ≈ 1.6) behaves as a genuinely independent feature. This necessitates dimensionality reduction before modeling.

VI. DIMENSIONALITY REDUCTION

A. Bartlett's Sphericity Test

Before applying PCA, Bartlett's Sphericity Test validated that the correlation matrix is significantly different from an identity matrix:

- $\chi^2 = 3330.94$, p-value $= 0.0$
- **Conclusion:** Reject the null hypothesis. Variables are correlated; PCA is valid.

All features were standardized using `StandardScaler` (zero mean, unit variance) before PCA and FA. Scaled feature matrix shape: $(243, 10)$.

B. Principal Component Analysis (PCA)

1) Eigenvalue Summary and Scree Plot

Table IV presents the full PCA eigenvalue summary. The scree plot (Fig. 10) shows that PC1 and PC2 clearly exceed the Kaiser threshold (eigenvalue > 1). PC3 is marginally below at 0.9209 but is retained because the cumulative variance at 3 components (82.27%) is substantially higher than at 2 components (73.10%), and PC3 has clear physical interpretation.

TABLE IV
PCA EIGENVALUE SUMMARY (FULL)

PC	Eigenvalue	Prop. Variance	Cumul. Variance
PC1	5.7505	57.27%	57.27%
PC2	1.5898	15.83%	73.10%
PC3	0.9209	9.17%	82.27%
PC4	0.7998	7.97%	90.24%
PC5	0.3930	3.91%	94.15%
PC6	0.2512	2.50%	96.65%
PC7	0.2232	2.22%	98.88%
PC8	0.0927	0.92%	99.80%
PC9	0.0162	0.16%	99.96%
PC10	0.0039	0.04%	100.00%

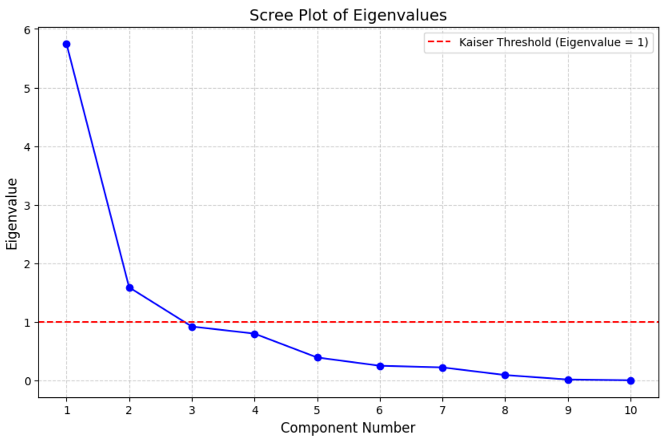

Fig. 10. Scree plot of PCA eigenvalues. Red dashed line = Kaiser threshold (eigenvalue = 1). Three components selected.

2) *Principal Component Loadings*

Table V and Fig. 11 present the eigenvector (loading) matrix for the 3 retained PCs.

TABLE V
PCA COMPONENT LOADINGS (EIGENVECTORS)

Feature	**PC1**	**PC2**	**PC3**
Temperature	+0.298	−0.348	+0.117
RH	−0.277	+0.399	−0.421
Ws	−0.040	+0.539	+0.420
Rain	−0.196	+0.215	+0.646
FFMC	+0.348	−0.232	−0.041
DMC	+0.374	+0.266	−0.089
DC	+0.330	+0.374	−0.249
ISI	+0.364	−0.073	+0.311
BUI	+0.370	+0.313	−0.145
FWI	+0.393	+0.119	+0.159

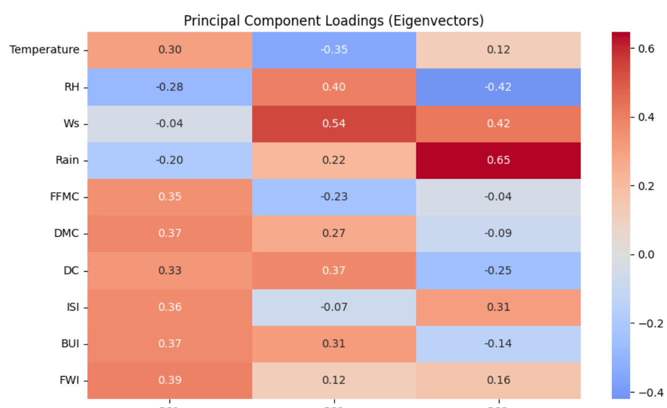

Fig. 11. Heatmap of PCA component loadings (eigenvectors) for PC1, PC2, and PC3.

The three components are physically interpreted as:

- **PC1 — Fire Danger/Dryness Index (57.3% variance):** Strong positive loadings on FWI (0.393), BUI (0.370), DMC (0.374), ISI (0.364), and FFMC (0.348); strong negative loading on RH (−0.277). Summarizes the overall heat-and-dryness fire danger state.
- **PC2 — Weather Shift Index (15.8% variance):** Dominated by Ws (+0.539), DC (+0.374), RH (+0.399), inversely related to Temperature (−0.348). Captures dynamic atmospheric transitions—cool, windy, humid air masses moving over historically dry terrain.
- **PC3 — Precipitation vs. Spread Trade-off (9.2% variance):** High Rain (+0.646), Ws (+0.420), ISI (+0.311), with negative RH (−0.421) and DC (−0.249). Represents acute rainstorm events where precipitation suppresses fire but wind simultaneously promotes spread.

Orthogonality verified: The correlation matrix of the 3 PCs shows all off-diagonal values exactly 0.0, confirming complete elimination of multicollinearity.

3) *PCA Biplot*

Fig. 12 shows the biplot of PC1 vs. PC2 with class coloring and feature loading arrows. Fire instances (red) cluster predominantly on the right (high PC1, high fire danger), while non-fire instances (blue) cluster on the left. The arrows confirm that RH and Rain point left (fire-suppressing direction) while FWI, BUI, DMC point right (fire-promoting direction).

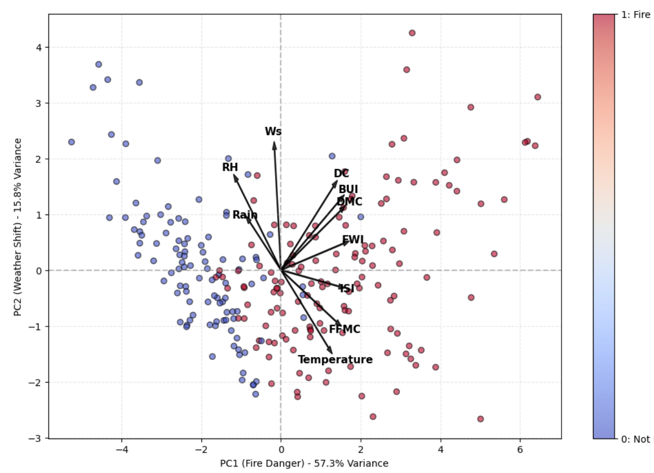

Fig. 12. PCA biplot: PC1 (Fire Danger, 57.3% variance) vs. PC2 (Weather Shift, 15.8% variance) with feature loading arrows. Red = Fire, Blue = Not Fire.

C. *Factor Analysis with Varimax Rotation*

Factor Analysis (FA) with 3 factors and Varimax rotation was applied to the standardized feature matrix. Unlike PCA (which maximizes explained variance), FA seeks interpretable latent constructs explaining the observed correlation structure.

1) *Factor Loadings*

Table VI and Fig. 13 present the Varimax-rotated factor loadings.

TABLE VI
FACTOR ANALYSIS LOADINGS (VARIMAX ROTATION, 3 FACTORS)

Feature	F1 (Dryness)	F2 (Atmospheric)	F3 (Spread)
Temperature	$+0.27$	$+0.51$	$+0.56$
RH	-0.11	-0.68	-0.42
Ws	$+0.096$	-0.012	-0.50
Rain	-0.25	-0.20	-0.38
FFMC	$+0.41$	$+0.57$	$+0.53$
DMC	$+0.88$	$+0.37$	$+0.10$
DC	$+0.93$	$+0.15$	$+0.064$
ISI	$+0.39$	$+0.91$	$+0.08$
BUI	$+0.95$	$+0.29$	$+0.088$
FWI	$+0.68$	$+0.72$	$+0.034$

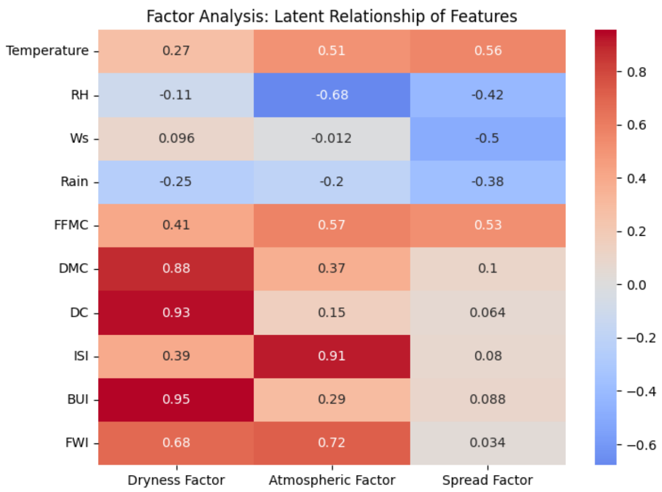

Fig. 13. Factor Analysis loadings heatmap (Varimax rotation). F1 = Dryness, F2 = Atmospheric, F3 = Spread.

The three factors are interpreted as:

- **Factor 1 — Dryness Factor:** Strongly loaded by BUI (0.95), DC (0.93), and DMC (0.88). Represents long-term accumulated fuel dryness—the foundational condition required for ignition.
- **Factor 2 — Atmospheric Factor:** Dominated by ISI (0.91), FWI (0.72), and FFMC (0.57), with negative RH (-0.68). Captures present-day atmospheric heat and low humidity that drives ignition intensity.
- **Factor 3 — Spread Factor:** Associated with Temperature (0.56), FFMC (0.53), and negative Ws (-0.50) and RH (-0.42). Governs fire propagation velocity once ignition has occurred.

2) FA Scatter Plot

Fig. 14 plots Factor 1 (Dryness) vs. Factor 2 (Atmospheric) colored by class. Fire instances cluster predominantly at higher Factor 1 (dryness) values, confirming that accumulated long-term fuel dryness is the most discriminative latent predictor.

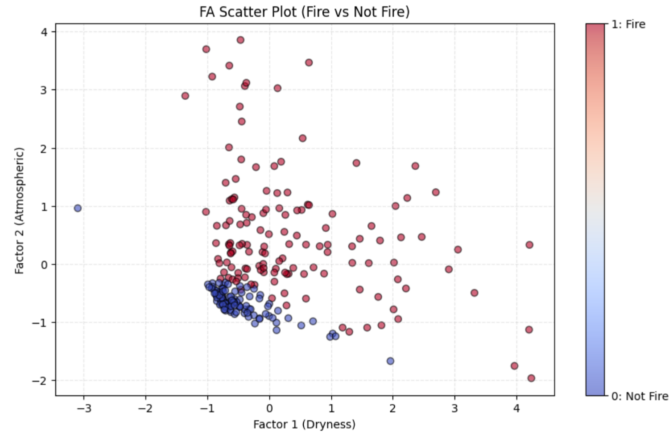

Fig. 14. FA scatter plot: Factor 1 (Dryness) vs. Factor 2 (Atmospheric), colored by fire class.

VII. CLASSIFICATION

A. Experimental Setup

Binary Logistic Regression (`max_iter = 1000`) is trained and evaluated on three feature variants:

- **Raw:** All 10 standardized features
- **PCA:** 3 principal components
- **FA:** 3 Varimax-rotated factors

All models use a 70:30 stratified train-test split (`random_state = 42`) and 5-fold cross-validation. Test set: 73 samples (29 Not Fire, 44 Fire).

B. Classification Results

1) Accuracy and Training Time

- **Raw Model:** Accuracy = 97.26%, Training Time = 0.0295 s
- **PCA Model:** Accuracy = 89.04%, Training Time = 0.0125 s ($\approx 2.4\times$ faster)
- **FA Model:** Accuracy = 97.26%, Training Time = 0.0413 s

2) 5-Fold Cross-Validation

- **PCA:** CV Accuracy = $88.05\% \pm 6.64\%$ (Interval: $[81.41\%, 94.69\%]$)
- **FA:** CV Accuracy = $95.48\% \pm 3.52\%$

3) Classification Reports

Table VII and Table VIII present the per-class classification reports for the PCA and FA models respectively.

TABLE VII
CLASSIFICATION REPORT: PCA MODEL (3 COMPONENTS)

Class	Precision	Recall	F1-Score	Support
0 (Not Fire)	0.86	0.86	0.86	29
1 (Fire)	0.91	0.91	0.91	44
Accuracy		0.89		73
Macro avg	0.89	0.89	0.89	73
Weighted avg	0.89	0.89	0.89	73

TABLE VIII
CLASSIFICATION REPORT: FA MODEL (3 FACTORS)

Class	Precision	Recall	F1-Score	Support
0 (Not Fire)	0.94	1.00	0.97	29
1 (Fire)	1.00	0.95	0.98	44
Accuracy		0.97		73
Macro avg	0.97	0.98	0.97	73
Weighted avg	0.97	0.97	0.97	73

4) ROC Curves and AUC

Fig. 15 shows the ROC curve for the PCA model (AUC $= 0.94$). Fig. 16 shows the ROC curve for the FA model (AUC $= 0.99$). The FA model's near-perfect AUC indicates that the latent factors—particularly the Dryness and Atmospheric factors—capture the class-discriminative structure more faithfully than PCA components.

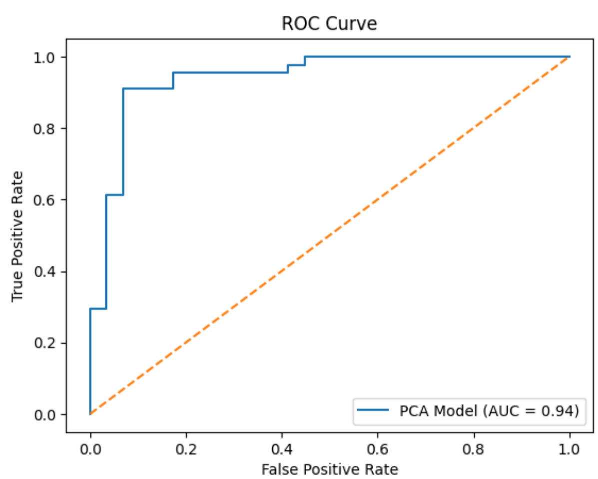

Fig. 15. ROC curve for PCA-based Logistic Regression model (AUC $= 0.94$).

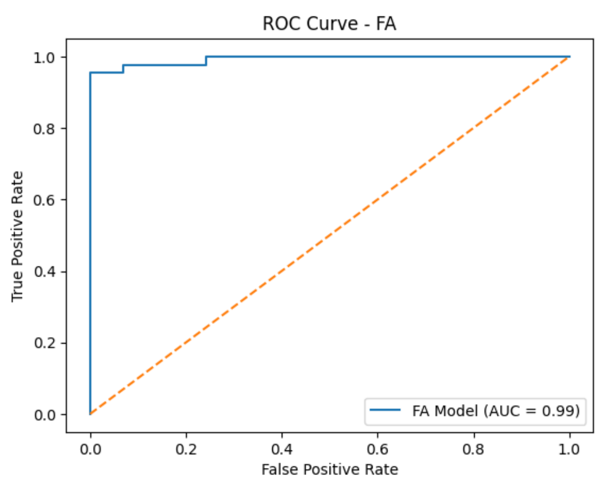

Fig. 16. ROC curve for FA-based Logistic Regression model (AUC $= 0.99$).

VIII. RESULTS AND DISCUSSION

A. Comparative Model Summary

Table IX consolidates all classification metrics across the three models.

TABLE IX
COMPARATIVE CLASSIFICATION RESULTS: RAW VS. PCA VS. FA

Model	Dims	Test Acc.	5-CV Acc.	AUC
Raw Features	10	97.26%	≈96.5%	≈0.99
PCA Components	3	89.04%	88.05% ± 6.64%	0.94
FA Factors	3	97.26%	95.48% ± 3.52%	0.99

B. Key Observations

Raw model instability: Although the raw model achieves the highest raw accuracy (97.26%), the confirmed severe multicollinearity (VIF $\gg$ 10 for BUI, DMC, DC) renders its logistic regression coefficients mathematically unstable and uninterpretable. The model is effectively using redundant information—feeding FWI, BUI, and DMC simultaneously is like feeding three copies of the same signal.

PCA trade-off: The PCA model achieves 89.04% accuracy with 70% fewer features and $\approx 2.4\times$ faster training. Multicollinearity is completely eliminated (all off-diagonal PC correlations = 0.0). The trade-off is a $\approx 8\%$ accuracy reduction and a slightly wider CV confidence interval (6.64% std), which reflects the information loss from variance-maximizing compression.

FA superiority: The FA model matches the raw model's accuracy (97.26%) while using only 3 dimensions, achieving a near-perfect AUC of 0.99 and a stable CV accuracy of 95.48% (std = 3.52%). Crucially, the FA factors are physically interpretable: Dryness (long-term fuel moisture), Atmospheric (current weather heat/dryness), and Spread (propagation conditions). This interpretability is directly actionable for fire management decisions.

Why FA outperforms PCA: PCA maximizes explained variance without regard for class labels. It may compress class-discriminative information into later components. FA, by contrast, identifies latent constructs that explain the correlation structure—constructs that directly correspond to fire physics. The Dryness factor (BUI, DC, DMC loadings ≥ 0.88) and Atmospheric factor (ISI = 0.91, FWI = 0.72) are exactly the dimensions that drive fire ignition and intensity.

Key fire predictors: FWI ($r = 0.72$), FFMC ($r = 0.77$), and ISI ($r = 0.74$) are the strongest individual correlates of fire class. RH ($r = -0.43$) and Rain ($r = -0.38$) are the strongest suppressors. The bivariate FFMC–Wind Speed analysis reveals that the danger threshold (FFMC > 80, Ws > 15 km/h) concentrates fire occurrences, confirming feature interaction as a critical modeling consideration.

Rain as a binary suppressor: Out of 110 rainy days, fires occurred on only 23 (20.9%), confirming rain's near-absolute suppression effect regardless of temperature. Among 133 rain-free days, fires were the majority class.

IX. Additional Insights from Data

Beyond standard exploratory analysis and modeling, several meaningful insights were derived to enhance interpretability and provide practical understanding of fire occurrence patterns.

A. Insight 1: Seasonal Fire Risk

Fire probability is highest during high-risk months (approximately 72.4%) and significantly lower during medium (41.7%) and low-risk periods (38.3%). This confirms a strong seasonal trend, where prolonged heat and dryness during late summer greatly increase fire occurrence.

B. Insight 2: Regional Fire Risk Variation

Region 1 exhibits a higher fire probability (approximately 64.5%) compared to Region 0 (approximately 48.4%). This indicates that regional climatic conditions significantly influence fire occurrence and highlights the importance of incorporating location-based features in predictive modeling.

C. Insight 3: Fire Risk Threshold Model

A high-risk zone was identified using simple thresholds (FFMC $>$ 80 and wind speed $>$ 15 km/h). The probability of fire occurrence in this zone is approximately 98.4%, indicating that extreme dryness combined with strong wind conditions almost always leads to fire. This demonstrates the effectiveness of simple rule-based approaches in identifying critical fire conditions.

D. Insight 4: Rain as a Fire Suppressor

Fire probability is significantly lower when rainfall is present (approximately 20.9%) compared to no-rain conditions (approximately 85.7%). This shows that rainfall acts as a strong suppressor by increasing moisture and reducing flammability, making it a key factor in fire prevention.

E. Insight 5: Composite Dryness Index

A composite Dryness Index was constructed by combining BUI, DMC, and DC to represent cumulative fuel dryness. The analysis shows that fire cases have significantly higher dryness values compared to non-fire cases. This confirms that long-term dryness is a major driver of fire occurrence and provides a simplified indicator of fire susceptibility.

F. Insight 6: Rule-Based vs Machine Learning

The rule-based model achieves moderate accuracy (approximately 68.3%), while the machine learning model achieves significantly higher accuracy (approximately 97.3%). This highlights that although rule-based methods are simple and interpretable, machine learning models are more effective in capturing complex relationships for accurate fire prediction.

X. Conclusion

This study presents a rigorous, end-to-end analytical pipeline for forest fire prediction using the Algerian Forest Fires Dataset. The pipeline covers region-based data splitting, cleaning, statistical regional validation via Welch's t-tests, comprehensive EDA across seven visualization types, formal multicollinearity diagnosis (VIF + Mahalanobis Q-Q), and dimensionality reduction using PCA and Factor Analysis.

The comparative classification study using Logistic Regression demonstrates:

- **Raw model (10-dim):** 97.26% accuracy but mathematically unstable due to extreme VIF values
- **PCA model (3-dim):** 89.04% accuracy, AUC = 0.94, zero multicollinearity, 2.4$\times$ faster training
- **FA model (3-dim):** 97.26% accuracy, AUC = 0.99, CV = 95.48%, physically interpretable factors (Dryness, Atmospheric, Spread)

Factor Analysis with Varimax rotation is the recommended approach for this dataset—it matches raw accuracy while providing stable generalization, complete multicollinearity elimination, and physically grounded feature representation directly applicable to fire risk management.

Future work should explore non-linear classifiers (Random Forest, XGBoost, SVM) applied to FA-transformed features, incorporation of spatial data, and real-time meteorological API integration for operational deployment as a fire risk alert system for Algerian authorities.

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